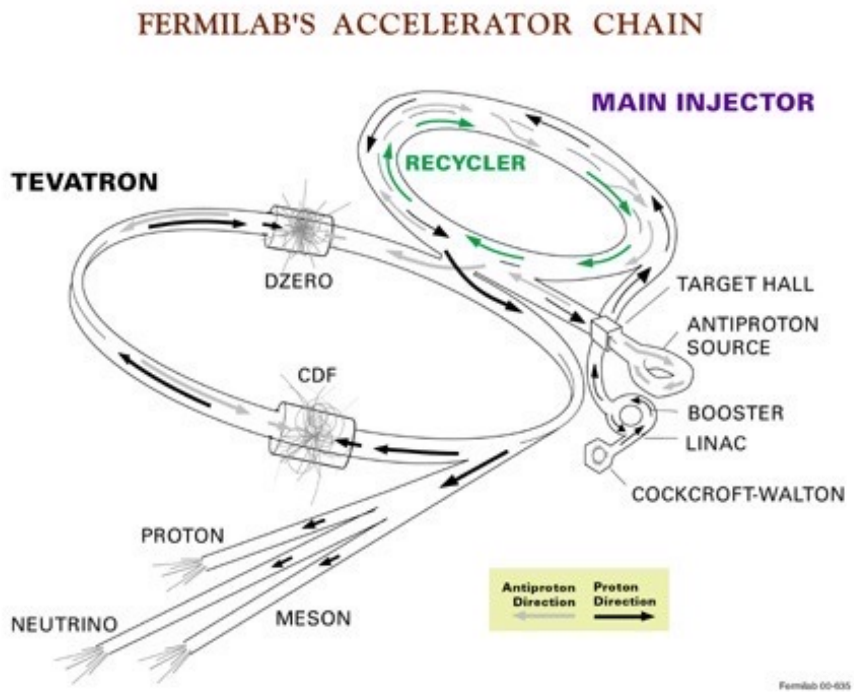
Discovery of top quark

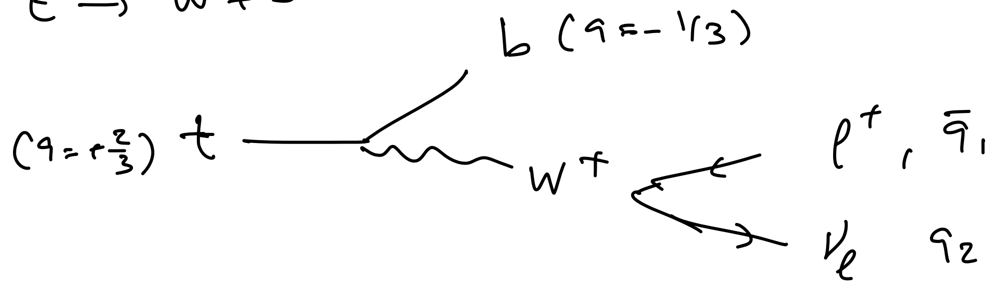
$p + \bar{p} \to t + \bar{t} + X$

Tevatron @ $\sqrt{s} = 1.96$ TeV

$p + p \to p + p + p + \bar{p}$



$t \to W + b$



W^+ DECAY MODES		Fraction (Γ_i/Γ)	Confidence level	p (MeV/c)
$\ell^+ \nu$	[b]	(10.86 ± 0.09) %		–
$e^+ \nu$		(10.71 ± 0.16) %		40189
$\mu^+ \nu$		(10.63 ± 0.15) %		40189
$\tau^+ \nu$		(11.38 ± 0.21) %		40170
hadrons		(67.41 ± 0.27) %		–

$p + \bar{p} \to t + \bar{t}$

$\bar{t} \to \bar{b} + W^-$, $W^- \to e^- \bar{\nu}_e$

$t \to b + W^+$, $W^+ \to e^+ + \nu_e$

Golden Channel

final state in the detector: $\underbrace{b + \bar{b}}_{\text{2 hadronic jets}} + \underbrace{\ell^+ + \ell^-}_{\text{OS leptons}} + \nu_{\ell_1} + \bar{\nu}_{\ell_2}$

Most events: $t + \bar{t}$

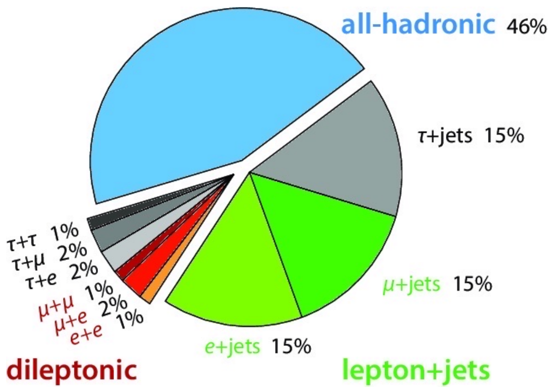
$\bar{t} \to \bar{b} + W^- \to q_1 \bar{q}_2$

$t \to b + W^+ \to q_3 \bar{q}_4$

Final state: $\underbrace{b + \bar{b}}_{\text{2 hadronic}} + \underbrace{q_1 + \bar{q}_2 + q_3 + \bar{q}_4}_{\text{4 hadronic jets from } u,d,s,c,b}$

6 jets

$t \to bW \approx 100\%$



e + jets:

$W^- \to e^- \bar{\nu}_e$

$W^+ \to q_1 \bar{q}_2$

$$\#(p\bar{p} \to b + \bar{b} + \ell_1^+ + \ell_2^- + \nu_{\ell_1} + \bar{\nu}_{\ell_2}) =$$

$$\sigma_{t\bar{t}} \times \mathcal{L}_{inst} \times \Delta t \times BF(t \to bW)\, BF(\bar{t} \to \bar{b}W) \times$$

$$\times BF(W \to \ell\nu) \times BF(W \to \ell\nu)$$

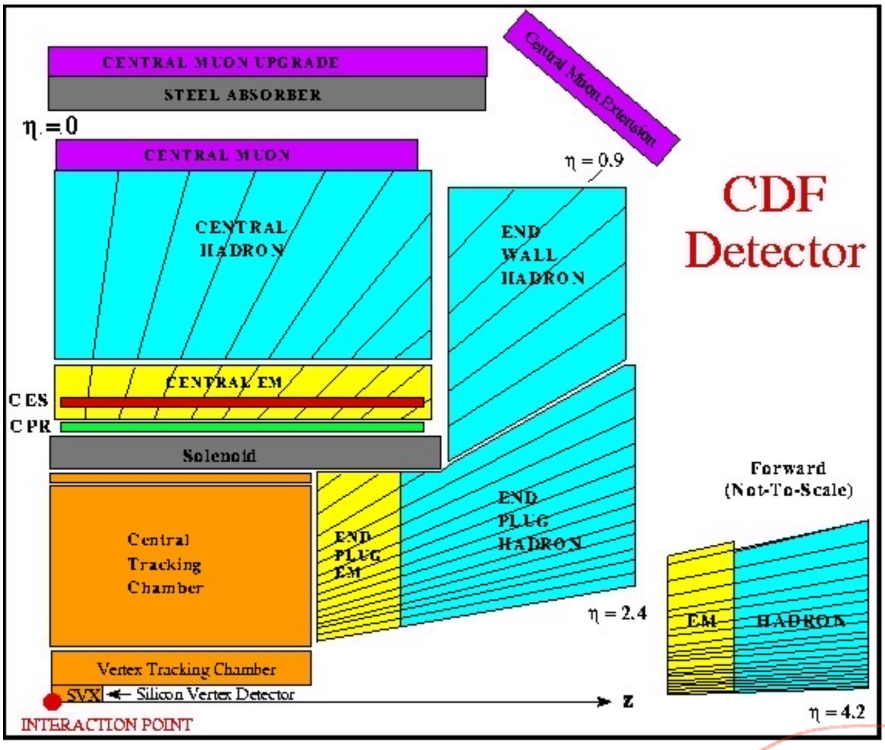
loose in statistics to gain in signal purity.

Golden mode:
- small # events
- best resolution
- better signal/bkg

tracker: for $\ell^\pm$ and distinguish + from −

EM Calorim. / Had Calorim.
- distinguish $e^\pm$ from hadrons
- measure hadronic jets

Muon system − distinguish $\mu^\pm$ from hadrons



$$t + \bar{t} \longrightarrow b + \bar{b} + \ell_1^+ + \ell_2^- + \nu_{e_1} + \bar{\nu}_2$$

how to measure their en/momentum?

$$t \longrightarrow b \; W^+$$
$$\hookrightarrow e^+ + \nu_e$$

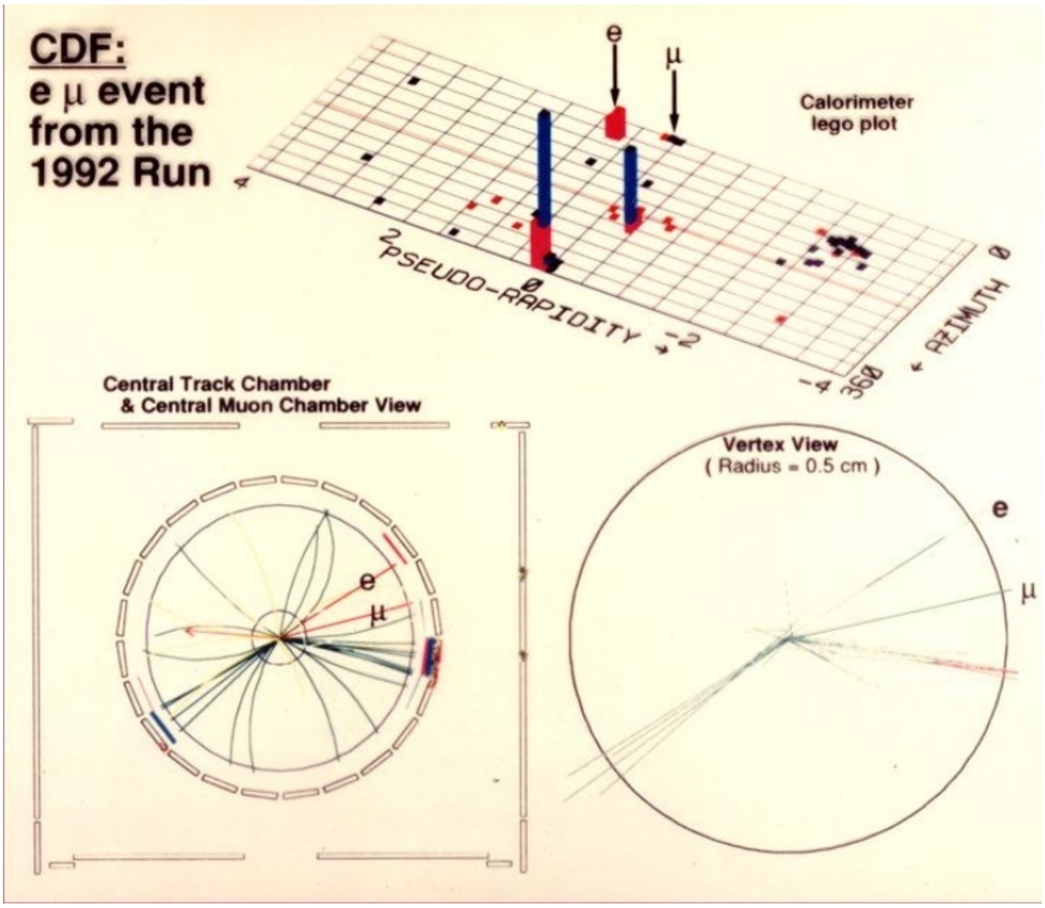
$$m_{inv}^2 = (\underline{P}_b + \underline{P}_e + \underline{P}_\nu)^2$$

$\hookrightarrow$?

neutrinos do not interact with detectors

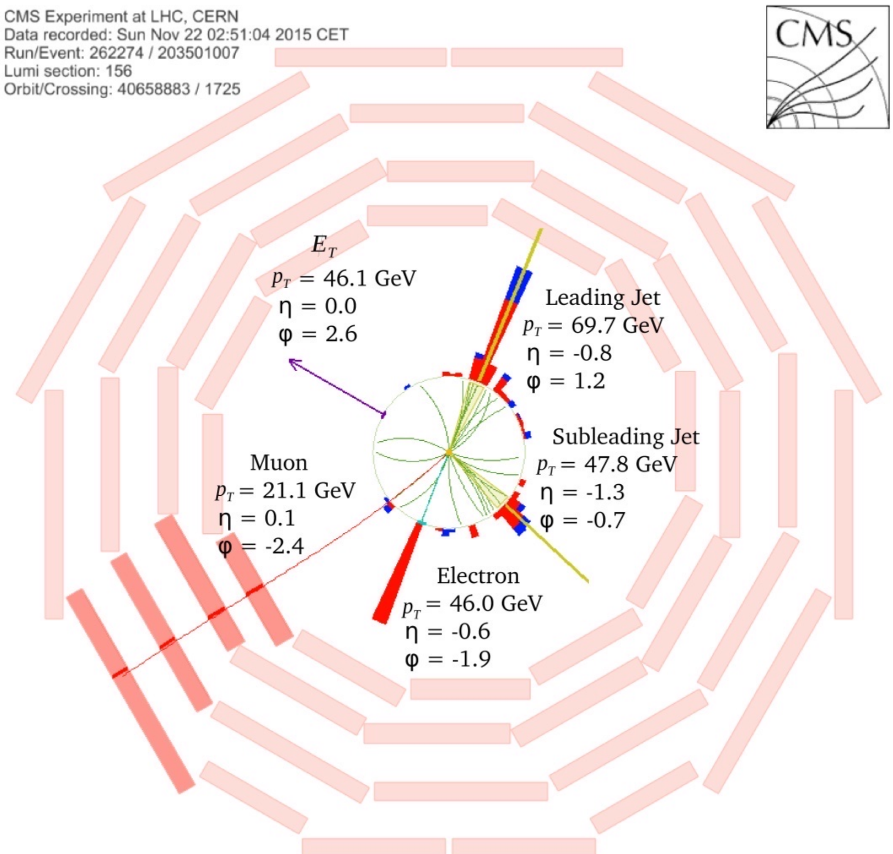
$\Rightarrow$ missing energy/momentum in the event

Exp Signature: $\underbrace{b + \bar{b}}_{\text{2 had jets}} + \underbrace{\ell^+ + \ell^-}_{\text{2 lepton}} +$ missing energy.



Blue:
had calor.
en. deposit

Red:
EM Calo
en. deposit



$\vec{P}_T = \vec{0}$ in initial state.

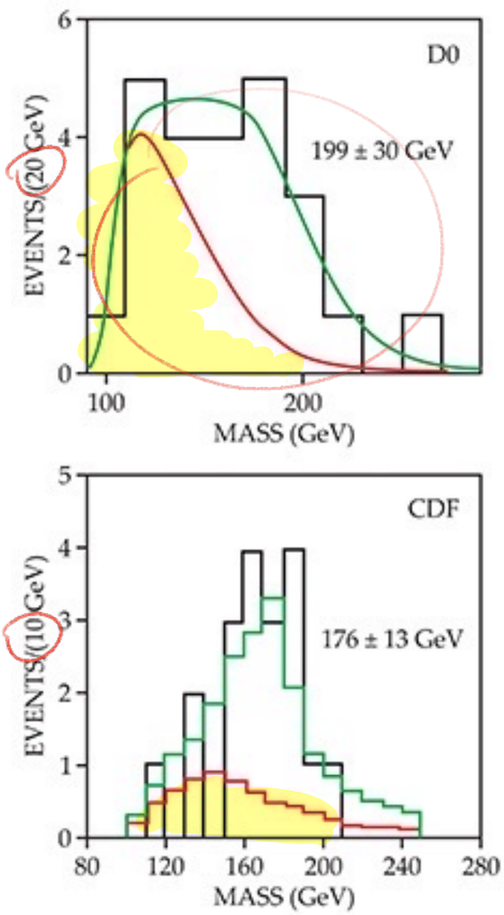
$\Rightarrow \quad \sum_i^{\text{final}} \vec{P}_i = \vec{0}$

$$\underbrace{\vec{p}_b^{\,T} + \vec{p}_{\bar b}^{\,T} + \vec{p}_{\ell^-}^{\,T} + \vec{p}_{\ell^+}^{\,T}}_{\text{measured.}} + \underbrace{\vec{p}_{\nu}^{\,T} + \vec{p}_{\bar\nu}^{\,T}}_{\substack{\text{not measured.} \\ \vec{p}_{T_{mis}}}} = \vec{0}$$

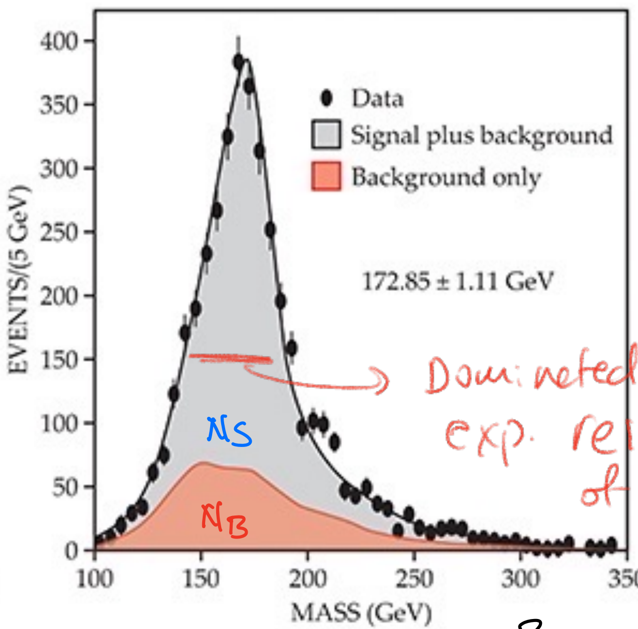
$$\sum_i \vec{p}_i^{\,T} = \vec{0} \quad \Rightarrow \quad \vec{p}^{\,T}_{mis} =: -\sum_i^{\text{measured}} \vec{p}_{T_i}$$

$$MET =: |\vec{p}^{\,T}_{mis}| = |\vec{p}^{\,T}_{\nu\bar\nu}|$$

$\Rightarrow$ look at invariant mass of $t \to b W$, $W \hookrightarrow \ell\nu$



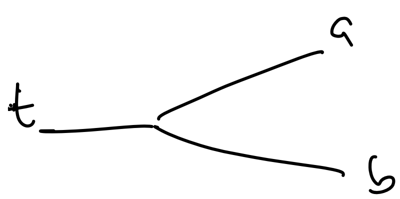
1994



Dominated by exp. resolution of jets

$$m_{inv}^2 = (P_b + P_\ell + P_\nu)^2$$

$$P = 0.3 \times B \times R$$



$$m_{ab} = (E_a + E_b)^2 - (\vec{p}_a + \vec{p}_b)^2$$

$\begin{pmatrix} u \\ d \end{pmatrix} \begin{pmatrix} c \\ s \end{pmatrix} \begin{pmatrix} t \\ b \end{pmatrix}$ $q = +2/3$, $q = -1/3$ Fermions.

So far no hint of additional quarks.

$\begin{pmatrix} e^- \\ \nu_e \end{pmatrix} \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix} \begin{pmatrix} \tau^- \\ \nu_\tau \end{pmatrix}$

τ^- → 1976 $e^+e^- \to \tau^+\tau^-$

ν_τ → 1999 @ Fermilab DONUT exp.

$\underbrace{1^{st} \quad 2^{nd} \quad 3^{rd}}_{\text{family of matter}}$

Exercise – 23.4.26

$\Gamma_i \to$ decay rate

$$\Gamma_i \propto |M|^2 \rho(\text{init} \to \text{fin})$$

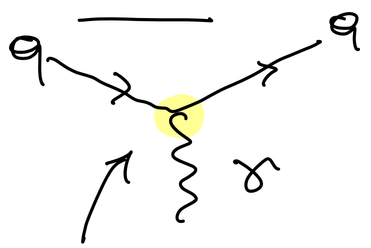
↳ kinematics

↳ dynamics

→ derive M w/ F.R.

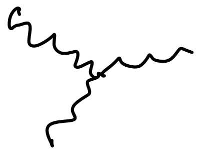
time →

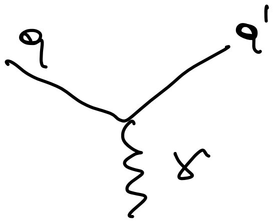
QED



$\sqrt{\alpha_{em}}$

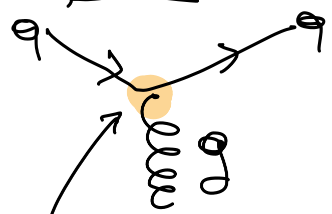
$$\alpha_{em} = \frac{(Ze)^2}{4\pi}$$

1)  NO X

2) 

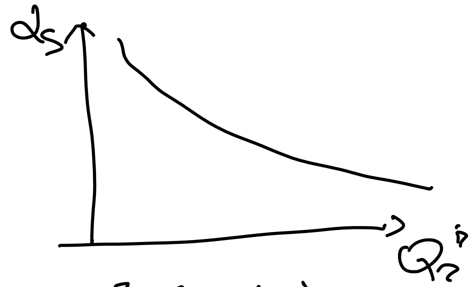
NO X

QCD

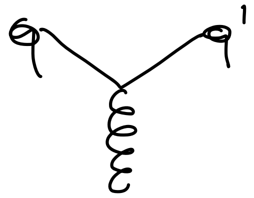


$\sqrt{\alpha_s}$

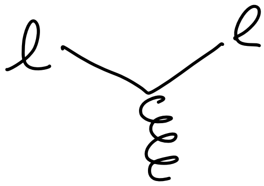
$$\alpha_s = \frac{g^2}{4\pi}$$



$Q^2 \nearrow \ \alpha_s \searrow$

1) 

NO X

2) 

NO X

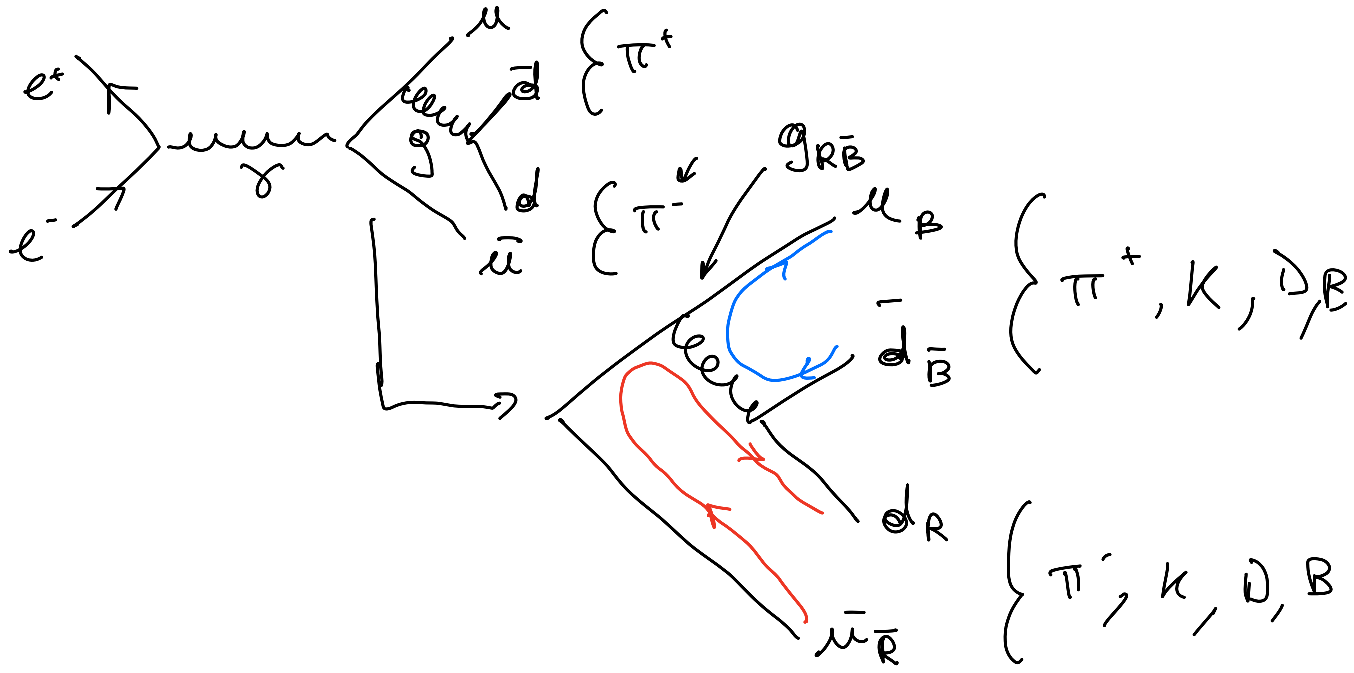
NB γ is 1^{--}

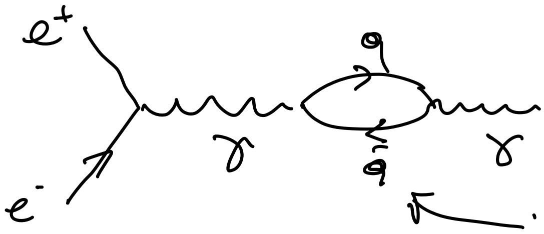
NB g is not eigenstate of C (b/c is colored)

$$C(gg) = (-1)^{c} \overset{c=0}{=} +1$$

$$C(ggg) = -(-1)^{c} = -1$$

ex $e^+e^- \to \pi^+\pi^-$





· resonance
· intermediate state

ρ^{0}

J/ψ, Υ

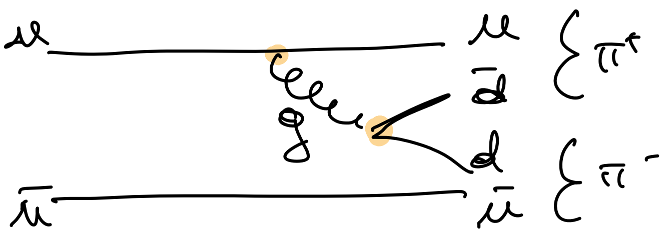
Vector mesons

	quarks	mass [MeV]	Γ [MeV]	Pref. decay
ρ	$\{u, d, \bar{u}, \bar{d}$	770	150	2π
ω		800	8	3π
ϕ	$s\bar{s}$	1020	4	KK
J/ψ	$c\bar{c}$	3100	0.09	$\{$hadrons
Υ	$b\bar{b}$	9500	0.05	

① Why Γ_ρ is large

$\rho \to \pi\pi$

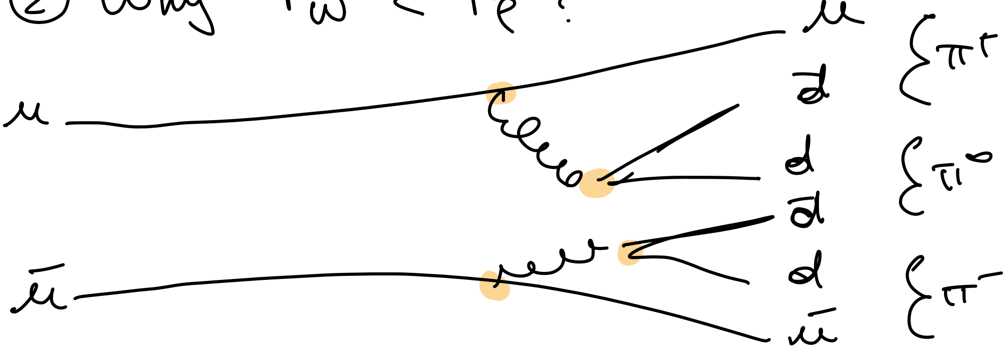
$\to$



1) $\Delta m = m_\rho - 2\, m_\pi \sim 500$ MeV $\Rightarrow$ large phase space

2) $M \propto \alpha_s$

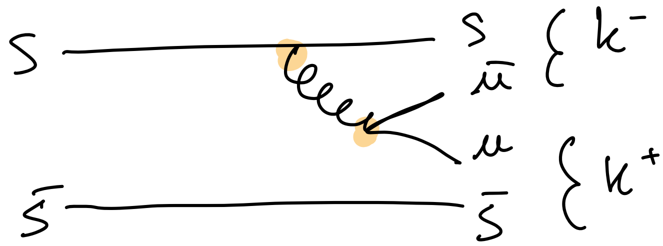
② Why $\Gamma_\omega < \Gamma_\rho$? $\omega \to 3\pi$



$M \propto \alpha_s^2$

$\to \Gamma_\omega < \Gamma_\rho$

③ Why $\Gamma_\phi < \Gamma_\rho$? $\phi \to K^- K^+$



$M \propto \alpha_s$

$\Delta m = m_\phi - 2 m_K = 1020 - 2\cdot(494) \sim 30 \text{ MeV}$

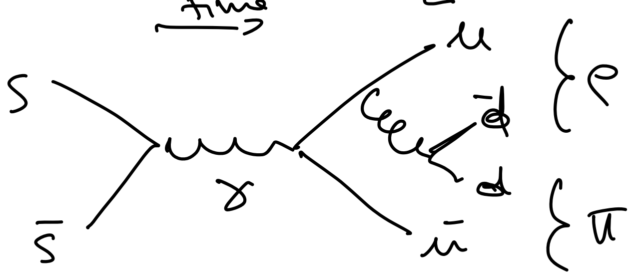
→ ④ Why $\phi \to \eta\gamma$ and $\phi \to \pi^0\gamma$ have very different BR?

$BR(\phi \to \eta\gamma) \sim 1\%$

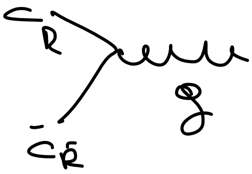
$BR(\phi \to \pi^0\gamma) \sim 0.01\%$

⑤ $\phi \to \rho\pi$ [$\phi \to \pi\pi$ forbidden $C(\phi) = -1$, $C(\pi\pi) = (-1)^2 = +1$]

time →



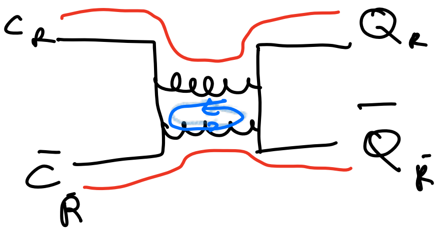
⑥ J/ψ STRONG DECAYS



NO × gluon is colored!

1 gluon

2 gluons NO ✗

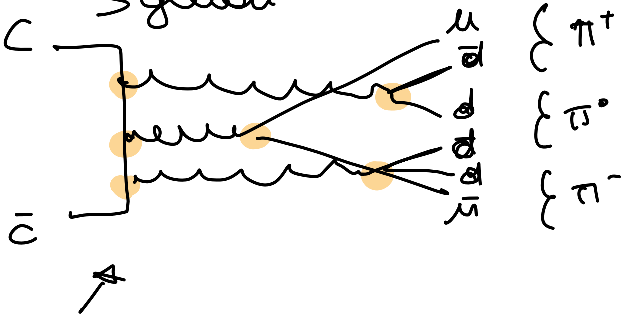


$J/\psi \to gg$?

$C(J/\psi) = -1$

$C(gg) = +1$

3 gluon



$J/\psi \to ggg$?

$C(ggg) = -1$ ✓

$M \propto \alpha_s^3$

$\hookrightarrow \Gamma(J/\psi) \ll$

ϕ vs J/ψ

$\phi \to KK$ ✓ $\quad J/\psi \to DD$ ✗ bc $m_{J/\psi} < 2 m_D$

OZI $\to$ processes w/ disconnected lines are suppressed

$\to$ 7 $\quad \phi \to 3\pi$ w/ OZI rule compare w/ $\omega \to 3\pi$